

DriftSense: Privacy-Preserving Intention-Aware Prediction and Model-Assisted Reflection for Browser-Based Digital Drift

[AUTHOR NAME], [Department / University], [Country]

Digital wellbeing tools commonly rely on domain blocking, screen-time limits, and usage dashboards. These approaches can misclassify purposeful browsing because the same website can support focused information seeking, an intentional break, or unplanned consumption. We present DriftSense, a privacy-preserving Chrome extension and local machine-learning pipeline for studying browser-based digital drift, defined as a participant-reported mismatch between a declared browsing intention and the session outcome. DriftSense records only configured-domain metadata, aggregate interaction counts, and short activity-window summaries; it does not collect page content, screenshots, messages, or keystroke values. We specify a two-phase field protocol with approximately 20 adult Chrome users. During a 10-day collection phase, participants declare an intention and provide a post-session reflection while no model-assisted prompt is shown. Time, domain, intention-only, activity-only, and intention-plus-activity models are then evaluated, and a three-minute model and intervention threshold are frozen. During a seven-day intervention phase, elevated-risk sessions are randomized with equal probability to a reflective prompt or a silent control, while both assignments retain the same intention and post-session questions. Prediction is evaluated with participant-aware and chronological validation, and the intervention is evaluated by comparing self-reported drift in randomized eligible sessions. Empirical collection has not yet occurred, so this manuscript reports the system and methodology without invented results. DriftSense studies self-reported intention alignment; it does not claim to detect attention, emotion, addiction, or mental health status.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning approaches*.

Additional Key Words and Phrases: digital wellbeing, browser extension, intention alignment, digital drift, self-report, privacy-preserving sensing, session-level prediction, reflective check-ins

1 Introduction

People often open distracting websites for legitimate reasons. A user may visit YouTube to find a tutorial, Reddit to answer a technical question, LinkedIn to check a message, or a news site to follow a specific event. The same sites can also support unplanned passive consumption or avoidance. This ambiguity creates a problem for common digital wellbeing tools: a domain category or elapsed time alone cannot reliably distinguish purposeful use from digital drift.

Existing tools often operationalize undesirable use through screen time, app or website category, or fixed blocking rules [3, 14, 17, 19]. Such mechanisms can be useful for self-control, but they can also generate false positives when a user has a valid reason to visit a monitored website. They may also generate false negatives when a short session is clearly misaligned with intention. Recent work has therefore shifted toward user agency, reflection, friction, and intention alignment. For example, self-nudging systems such as *one sec* introduce a short pause before app access [9]; Purpose Mode reduces distracting interface patterns while preserving access to social media websites [12]; and PauseNow frames digital wellbeing as alignment between intentions and behavior [21].

We use the term *browser-based digital drift* to describe a session-level intention-behavior mismatch: the user begins with one declared purpose but later reports that the visit did not match that intention. This framing is deliberately narrower than addiction, attention detection, or productivity optimization. DriftSense does not infer clinical status or inner mental state. Instead, it collects explicit intention and explicit post-session reflection, then evaluates whether those self-reported labels can be predicted from lightweight browser-session signals.

Author's Contact Information: [Author Name], [Department / University], [City], [Country], [Email].

DriftSense is designed around a local-first Chrome Manifest V3 extension and a reproducible machine-learning pipeline. Before accessing a configured domain, the user declares an intended activity using a neutral set of options: *work or study task, learning or tutorial, specific information, communication or community, planned entertainment or break, open-ended browsing, or opened accidentally*. The categories avoid assuming that work is inherently aligned or leisure is inherently drift. During the session, the extension records duration, click count, scroll count, idle time, focus changes, and short activity-window summaries. After the session, it asks whether the visit matched the original intention. In the intervention phase, a locally computed three-minute risk estimate determines eligibility for a session-level randomization: half of elevated-risk sessions receive a brief reflective prompt and half form a silent control. The website remains accessible in either assignment.

The central research questions are:

RQ1: Can declared browsing intention combined with lightweight browser activity signals predict self-reported session-level drift better than time-based or domain-based approaches?

RQ2: Among sessions classified as elevated risk after three minutes, does a model-assisted reflective prompt reduce subsequent self-reported drift compared with a randomized silent control?

The expected contribution is an implementation and evaluation framework for intention-aware drift prediction and lightweight reflective support in browser sessions. The contribution is not activity tracking, dashboarding, or blocking alone. It is the combination of privacy-preserving activity signals, intention-grounded self-report labels, explicit comparisons against weak baselines, early prediction without future-session leakage, and a session-level randomized test of model-assisted reflection. The 10-day collection phase is used for model development; it is not treated as a causal control for the later intervention phase.

2 Related Work

2.1 Digital Wellbeing and Digital Self-Control Tools

Digital wellbeing research has documented a broad ecosystem of tools that help users manage technology use through blocking, friction, monitoring, goal setting, reminders, interface changes, and reflection [3, 14, 17, 19]. Roffarello and De Russis’ systematic review and meta-analysis shows that digital self-control tools vary widely in intervention strategy and evaluation quality, and that many tools emphasize reducing overuse rather than supporting more contextual judgments about why use occurs [17]. Plackett et al.’s journal review of social media use interventions similarly found mixed evidence for limiting-use and abstinence interventions, reinforcing the need to evaluate more contextual approaches rather than assuming that less use is always the appropriate target [19]. Biedermann et al. review digital self-control interventions for distracting media multitasking and identify a mixture of blockers, goal-setting tools, visualizations, and awareness mechanisms [3].

This body of work motivates DriftSense in two ways. First, it shows that time and blocking are common operational levers. Second, it reveals an opportunity for session-level interpretation that treats user intention as a first-class input rather than a background assumption. DriftSense therefore evaluates time and domain baselines, but it does not treat either as sufficient.

2.2 Browser and Social Media Distraction Interventions

Prior HCI systems have explored interventions that change access conditions or interface affordances. Lyngs et al. reviewed digital self-control tools through dual-systems theory and described mechanisms such as commitment,

blocking, reminders, and self-monitoring [14]. Later design interventions for social media explored ways to help users maintain self-control without simply removing access [15].

Recent work has focused more directly on platform design patterns. Purpose Mode lets users toggle attention-capturing patterns on social media websites and reports reduced perceived distraction in a field deployment [12]. Research on attention-capture deceptive designs also argues that interface patterns such as infinite scroll and autoplay can shape attentional harms [18]. DriftSense differs from such interface-modification work because it does not remove or alter social media features. Instead, it studies whether session-level drift can be predicted from declared intention and lightweight activity signals, and whether a reflective check-in can help users reassess sessions at risk of drift.

2.3 Intention-Aware and Reflective Interventions

Intention-aware systems directly ask users to articulate goals or create moments of reflection before use. The *one sec* app inserts a short pause before opening selected apps; its observational field deployment reported fewer opening attempts over time, while a preregistered online experiment provided stronger evidence for the mechanisms of friction and an explicit skip choice [9]. PauseNow proposes an intention-alignment framework, but it does not report a deployed outcome evaluation [21]. StayFocused compared a timer, reflective prompts, and chatbot support; several behavioral differences were marginal or nonsignificant, although the chatbot condition was more likely to return after an attempted exit [13]. MindShift reported higher intervention acceptance for LLM-generated messages than a fixed reminder, but behavioral reductions were mixed and its richer mental-state data create a different privacy boundary from DriftSense [20].

DriftSense borrows the idea that intentions matter, but it uses that idea for measurement, prediction, and lightweight reflective check-ins rather than website blocking or mental-state inference. The pre-session prompt is not meant to block access. The post-session reflection supplies a self-reported session label. This allows DriftSense to evaluate whether intention adds predictive value beyond domain and time, and whether model-triggered reflection can reduce self-reported drift without changing website interfaces.

2.4 Personal Informatics and Activity Tracking

Personal informatics systems help users collect and reflect on data about their behavior. In digital wellbeing, activity summaries and dashboards can make technology use visible, but reflection can be more useful when the data is connected to meaningful goals rather than abstract totals. WellScreen, for example, scaffolds daily reflection by asking people to estimate and report smartphone use, emphasizing self-awareness beyond raw screen time [2]. Work on multi-device digital wellbeing also shows that users’ experiences span devices and contexts, complicating simplistic time-based interpretations [16].

DriftSense uses dashboarding and export features only as supporting infrastructure. The dashboard summarizes configured-domain sessions, drift sessions, non-drift sessions, top drift domains, intention distributions, duration, activity patterns, and trends. These summaries help participants and researchers inspect the data, but the paper’s central object of evaluation is the drift classifier.

2.5 Regret, Intention Mismatch, and Session-Level Modeling

Regretful or meaningless technology use is closely related to digital drift. Cho et al. studied regretful smartphone use at the app-feature level and showed that regret depends on the kind of activity, not merely the amount of time spent [7]. Guo et al. analyzed a large corpus of smartphone screenshots with multimodal LLMs and found that regret varies by

user intention and type of social media use [10]. A recent seven-day preprint with 21 participants found that perceived intention–use mismatch related more strongly to regret than duration did. Its phone-context CatBoost model achieved mean within-person AUC of 0.732, but leave-one-participant-out AUC was only 0.536, demonstrating the need to separate known-user and unseen-user performance [1].

These studies motivate DriftSense’s label definition. A drift label is not assigned because a session is long or because the domain is social media. A drift label is assigned when the user explicitly reports that the visit did not match the original intention. This choice accepts that labels are subjective and potentially noisy, but it makes the target construct closer to the experience of drift than a generic time threshold.

2.6 Privacy and Ethical Concerns in Behavioral Tracking

Behavioral tracking creates privacy risks, especially when systems collect fine-grained content, messages, screenshots, or inferred mental states. Digital wellbeing scholarship has argued that tools should respect user agency and avoid framing all technology use as harmful [4, 5, 21]. Digital phenotyping reviews also highlight the need to distinguish behavioral measurement from clinical inference and to avoid overclaiming from sensor data [11].

DriftSense responds by minimizing data collection. The prototype records only monitored domains and metadata needed for session-level modeling. It does not collect page text, passwords, screenshots, messages, full history, source code, keystroke content, webcam data, face identity, or emotion data. The extension stores data locally first and provides export and delete-all-data controls. These constraints reduce model expressiveness, but they keep the prototype aligned with the paper’s narrow claim.

2.7 Synthesis and Research Gap

Table 1 compares the closest systems along the dimensions that define the DriftSense gap. The timing of support ranges from pre-access friction to post-use reflection, but intention is usually either absent, collected after use, or used to generate an intervention rather than a session label. PauseNow is the closest conceptual framing but has no outcome evaluation; *Before You Scroll Again* is the closest predictive study but predicts regret from pre-session phone and wearable context, with intention–use mismatch reported after the session rather than declared before it [1, 21].

Across this literature, interventions operate before access through friction, during use through interface modification or contextual prompting, and after use through reflection. However, the selected work does not evaluate the exact combination proposed here: a declared pre-session browser intention, content-free early activity signals, a post-session intention-alignment label, explicit time/domain/intention/activity baselines, and a randomized model-assisted prompt. Evaluation quality also motivates the protocol. Purpose Mode used a fixed baseline-then-intervention order, the *one sec* field deployment lacked a concurrent control, and systematic reviews identify short duration, missing controls, homogeneous convenience samples, and nonstandard outcomes as recurring weaknesses [3, 9, 12, 17, 19]. One review found that only 7 of 35 between-subject digital self-control studies used a no-intervention control and that the median evaluation lasted 21 days [17]. DriftSense cannot solve the short follow-up or population-diversity problems in a small pilot, but session-level randomization among equally eligible sessions provides a stronger prompt comparison than a simple before–after design.

Table 1. Comparison of the selected systems most directly informing DriftSense. “Randomized contrast” means that the stated intervention mechanism was compared under random assignment, not merely that a study contained multiple phases.

Work	Intervention timing	Role of intention	Primary evidence	Predictive evaluation	Randomized contrast
Purpose Mode [12]	During use; interface features toggled	Goal alignment measured in EMA	Two-week field deployment; perceived distraction and use time	No session classifier	No; baseline always preceded intervention
MindShift [20]	At self-identified habitual app use	Purpose and reported mental state condition LLM messages	Formative studies and five-week field experiment	No drift classifier	LLM order counterbalanced; baseline fixed first
StayFocused [13]	Attempted exit from a focus period	User initiates a focus goal	Five-week, three-group field study	No session classifier	Between-group versions; several behavioral results mixed
PauseNow [21]	Proposed across recognition, intervention, and reflection	Original intention is central	Conceptual framework	Proposed, not validated	No outcome study
one sec [9]	Before app access	Reason for opening is elicited around the nudge	Six-week observational field data plus online mechanism experiment	No session classifier	Online experiment only; no field control
Before You Scroll Again [1]	Predicts before a social-media session	Intention–use gap reported after use	Seven-day sensing and experience-sampling study	CatBoost; within-person and participant-held-out tests	No intervention test
WellScreen [2]	Daily post-use reflection	Estimated use compared with actual use	Two-week mixed-methods technology-probe deployment	No session classifier	No control condition
DriftSense (planned)	Three minutes into an elevated-risk browser session	Declared before use; revisited after use	Ten-day collection plus seven-day intervention	Time, domain, intention, and activity baselines; early and participant-aware tests	1:1 prompt versus silent control among eligible sessions

3 System Design

3.1 Evidence-Derived Design Requirements

DriftSense is designed for a research setting where participants voluntarily install a browser extension, configure monitored domains, and export a local dataset for analysis. Five requirements follow from the reviewed evidence:

- (1) **Measure alignment, not assumed harmfulness.** Screen time and domain are retained as baselines, but the target label must come from the participant’s declared purpose and later reflection. This follows evidence that goal alignment and intention–use mismatch can be more informative than duration alone [1, 12].

- (2) **Intervene lightly and preserve agency.** Reflection should provide an explicit choice and never block the site. A single neutral check-in follows evidence that friction and an opt-out can disrupt automatic action, whereas repetitive or inflexible prompts can be irritating [9, 13].
- (3) **Minimize and localize sensing.** The system should use aggregate interaction counts and hostname-level context while excluding content, screenshots, messages, and key values. Local storage, inspection, export, and deletion must be participant controlled.
- (4) **Keep measurement stable across conditions.** The same intention and post-session questions should be used throughout collection and intervention so that the mid-session prompt is the only randomized interface difference.
- (5) **Make prediction and intervention auditable.** Exported records should support weak baselines, one-, three-, and five-minute prediction, participant-aware validation, a frozen deployment model, and verification of every eligibility and randomization decision.

Figure 1 summarizes the planned DriftSense extension and local machine-learning pipeline.

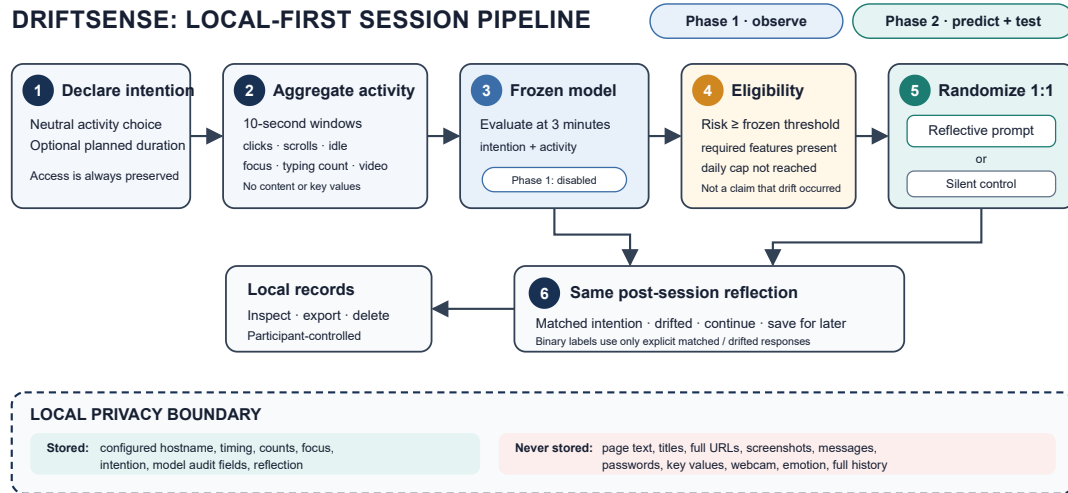


Fig. 1. DriftSense architecture and intervention decision flow. The prediction and randomization steps are disabled during the 10-day collection phase and enabled with a frozen model during the seven-day intervention phase.

3.2 Chrome Extension Architecture

The collection prototype uses Chrome Manifest V3 with React and TypeScript. It contains a background service worker for session monitoring, content scripts for aggregate activity counting on monitored domains, a popup for quick status, an options page for domain configuration and privacy controls, and a dashboard for summaries and export. The intervention build will add a fixed local model evaluated after three minutes, an elevated-risk threshold, a 1:1 session-level randomizer, and a reflective prompt. No remote inference service is required.

3.3 Monitored Domains and Study Phases

Users configure domains that they consider drift-prone or worth studying. Default examples include `youtube.com`, `facebook.com`, `reddit.com`, `instagram.com`, `x.com`, `linkedin.com`, news websites, and custom domains. DriftSense only tracks configured domains.

DriftSense supports two consecutive phases for the same cohort:

- (1) **Collection phase.** For 10 days, the extension asks for intention, records lightweight signals, and asks the post-session reflection. No model-assisted mid-session prompt is shown.
- (2) **Intervention phase.** For seven days, the same measurement flow remains active. At three minutes, the frozen model evaluates the session. Elevated-risk sessions are randomized with probability 0.5 to a reflective prompt or a silent control.

In both phases, when the user opens a monitored domain, the extension displays the same pre-session intention prompt:

Why are you opening this?

The response options are:

- Work or study task
- Learning or tutorial
- Specific information
- Communication or community
- Planned entertainment or break
- Open-ended browsing
- Opened accidentally

These choices describe the participant's intended activity rather than labeling an activity as productive or problematic. After selection, the site remains accessible. The prompt is a labeling and reflection mechanism, not a blocker.

In the intervention phase, an elevated-risk session assigned to the prompt arm receives a neutral reflective check-in such as:

Still here for your original reason?

The responses are *Continue intentionally* and *Finish now*. The prompt does not display the probability or claim that drift occurred; this reduces demand and avoids presenting a model estimate as a fact. Silent-control sessions receive no mid-session interface. Both assignments retain the same post-session label question. The intervention never blocks the website and appears at most once per session, with a prespecified daily cap to limit burden.

3.4 Session Tracking

A session begins when the user opens or focuses a monitored domain. The pre-session intention answer is associated with that session. A session ends when the tab is closed, the monitored domain is left, a timeout condition is met, or the user completes the post-session reflection. The primary participant CSV records the following 13 fields:

- `session_id`
- `participant_id`
- `start_time`
- `domain`

- declared_intention
- intended_duration_minutes
- duration_seconds
- click_count
- scroll_count
- keyboard_activity_count, without key values
- idle_seconds
- focus_loss_count
- drift_label

The intervention build adds a separate allowlisted log keyed by `session_id`. It records the model version and prediction offset; probability and threshold; eligibility and randomized assignment; prompt display and response; and any suppression reason such as reaching the daily cap. Separating this log preserves compatibility with the primary participant CSV and makes the randomized comparison auditable. Exact field names are documented in the export schema.

3.5 Activity Windows

Every 10 seconds, the content script records an activity-window summary. Each window contains:

- session_id
- timestamp_offset_seconds
- clicks_in_window
- scroll_events_in_window
- keyboard_activity_in_window, without key values
- idle_in_window
- tab_focused
- video_playing
- url_domain_only

These windows support first-3-minute prediction and full-session modeling. They also preserve the privacy boundary: no page text, screenshot pixels, input content, private messages, full browsing history, or source code are collected.

3.6 Post-Session Reflection and Labeling

After a configurable duration or session boundary, DriftSense asks the same post-session reflection question in both study phases and both randomized intervention assignments:

Did this visit match your original intention?

The response options are:

- Yes
- No, I drifted
- Continue intentionally
- Save for later

The primary binary labels are:

FROM DECLARED PURPOSE TO A SESSION-LEVEL LABEL

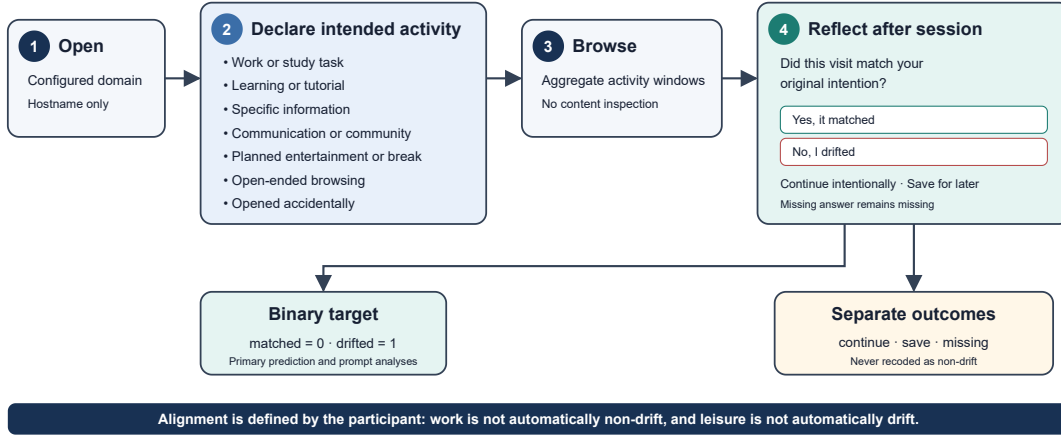


Fig. 2. DriftSense session-labeling flow. Both phases use the same intention declaration and post-session reflection so that the label definition remains stable.

$$y = \begin{cases} 0, & \text{if answer is Yes} \\ 1, & \text{if answer is No, I drifted} \end{cases} \quad (1)$$

Continue intentionally and *Save for later* are retained as non-binary outcomes and excluded from the primary binary classification and intervention analyses. Missing answers also remain unlabeled; neither non-binary nor missing outcomes are recoded as non-drift.

Figure 2 illustrates the session-labeling flow from monitored-domain access through intention declaration, activity-window logging, and post-session reflection.

3.7 Dashboard and Export

The participant-facing dashboard is a local reflection and transparency interface organized into five views. An *Overview* reports monitored and labeled sessions, self-reported drift rate, average duration, and monitoring status. *Trends* visualizes delayed aggregate patterns. *Intention Alignment* compares intention categories, domains, and post-session outcomes. *Session History* exposes the allowlisted metadata for individual records, and *Data and Privacy* supports export and deletion. These views use descriptive language and do not infer productivity, attention, addiction, or mental state.

The intervention build will add a compact summary of randomized prompt events and participant responses. Risk probabilities remain researcher-facing audit data rather than claims shown to participants. A session-detail view may show prompt assignment and response, but never page titles, full URLs, page text, screenshots, messages, or keystroke values. The data view allows participants to pause monitoring, configure domains, inspect field definitions, delete individual sessions, delete all local data, and export:

- a primary participant CSV named from the anonymous code, such as P01.csv;
- an optional activity-windows CSV for early-prediction analysis; and
- an optional complete JSON audit bundle containing allowlisted session and activity-window records.

Participant CSV files share the same 13-column modeling schema and are combined locally into `data.csv`. Missing or non-binary reflection outcomes remain blank in the participant source files and are not silently recoded as non-drift.

All dashboard queries run over locally stored records. The dashboard remains available in both phases and updates only after sessions or at a delayed aggregate interval, so it does not become an unplanned real-time intervention. Missing and uncertain labels remain visible and are not counted as non-drift. Dashboard-opening behavior is not logged. The dashboard supports transparency and data inspection; it is not treated as the paper’s novelty claim.

4 Drift Prediction and Model-Assisted Reflection

4.1 Prediction Task

The main task is binary session-level prediction:

Given declared intention, domain metadata, and lightweight activity signals from a prompted browser session, predict whether the user will later report that the session was drift.

The positive class is the self-reported *No, I drifted* label. The negative class is the self-reported *Yes* label. Sessions labeled *Continue intentionally* or *Save for later* are analyzed separately or excluded from the binary task. The resulting model outputs a session-level drift probability. During Phase 2, crossing the frozen threshold determines eligibility for prompt-versus-silent-control randomization; the probability is not treated as attention detection.

4.2 Feature Groups

The tabular feature set includes:

- declared intention
- domain category
- time of day
- day of week
- duration so far
- total clicks
- total scrolls
- total keyboard activity count
- idle ratio
- active ratio
- video ratio
- focus loss count
- tab switch count
- average activity per minute
- first-3-minute activity summaries

Activity windows are truncated at one, three, and five minutes to construct early feature sets. The three-minute set is used by the deployed intervention model, while one-minute, five-minute, and full-session sets are offline comparisons. Every early feature is computed only from observations available by its cutoff; final duration or later activity is never used for an earlier prediction.

Figure 3 shows the planned preprocessing, modeling, and evaluation pipeline for tabular drift prediction and model-assisted reflection.

TWO PHASES, TWO DISTINCT EVALUATION QUESTIONS

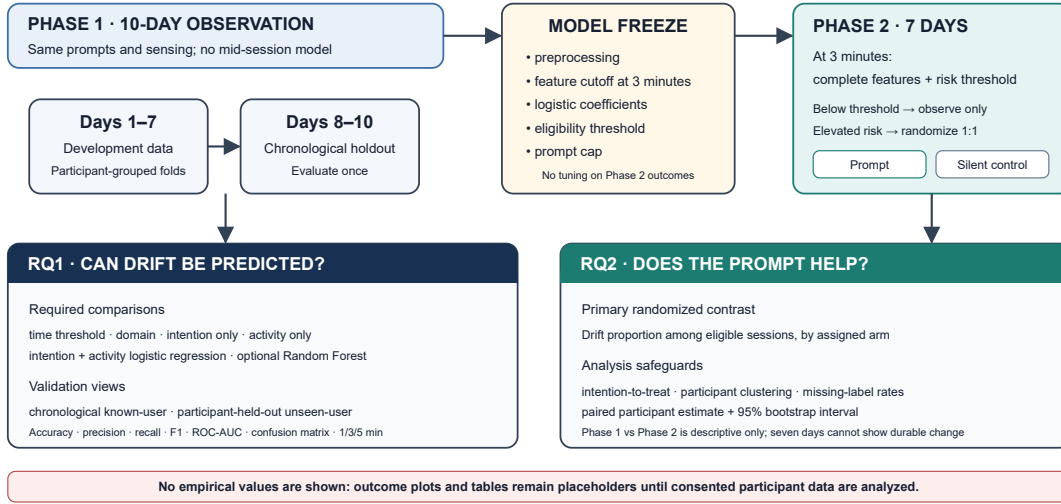


Fig. 3. Planned study and evaluation pipeline. RQ1 evaluates prediction using Phase 1 only; RQ2 evaluates the prompt using randomized eligible Phase 2 sessions. The Phase 1–Phase 2 change is not the causal intervention contrast.

4.3 Baselines

DriftSense evaluates four baselines before the main models:

- (1) **Time-threshold baseline.** Predict drift if session duration exceeds a threshold tuned on the validation set.
- (2) **Domain baseline.** Estimate each domain’s drift tendency from training data only and classify validation or test sessions using the frozen training-derived rule.
- (3) **Intention-only logistic regression.** Predict from declared intention and basic context features only.
- (4) **Activity-only logistic regression.** Predict from aggregate activity signals without declared intention.

These baselines test whether the proposed intention-plus-activity models add value beyond common digital wellbeing heuristics.

4.4 Main Models

The primary model is intention-plus-activity logistic regression. It is selected for interpretability, modest data requirements, and straightforward local deployment as a fixed preprocessing specification, coefficient vector, intercept, and threshold. A Random Forest provides an optional nonlinear robustness comparison if the number and class balance of labeled sessions are adequate. CatBoost or XGBoost may be reported only as secondary offline models [6, 8]; they are not required for intervention deployment. Deep sequence models are outside the first study because 15–20 participants are unlikely to provide enough independent data to justify them.

4.5 Early Prediction

Early prediction evaluates whether drift can be estimated before the full session ends. The offline comparison evaluates:

- first 1 minute
- first 3 minutes
- first 5 minutes
- full session

The three-minute model is chosen in advance for intervention deployment because it allows an early but non-immediate reflection point. Model family, preprocessing, threshold, and prompt cap are frozen before the intervention phase. Exceeding the threshold makes a session eligible for randomization; it does not automatically show a prompt. The model and randomization run locally, and the check-in supports reassessment rather than blocking access.

5 Methodology

5.1 Study Design

We plan a 17-day, two-phase, in-the-wild pilot with one cohort. Phase 1 is a 10-day observational collection period with consistent pre-session intention and post-session reflection prompts but no model-assisted mid-session intervention. After a short model-development interval, Phase 2 is a seven-day intervention period using the frozen model. In Phase 2, sessions that remain active for three minutes and exceed the frozen risk threshold are randomized 1:1 to a reflective prompt or a silent control. This micro-randomized comparison is the primary basis for RQ2. A simple before–after comparison between Phase 1 and Phase 2 will be reported only descriptively because phase order, novelty, and temporal changes prevent causal attribution.

The target is 20 completers; 22–24 participants may be recruited to allow for attrition. Feasibility targets are at least 500 complete binary-labeled Phase 1 sessions overall, at least 20 labeled Phase 1 sessions per participant where browsing frequency permits, and at least 100 randomized elevated-risk Phase 2 sessions. These are operational targets rather than a formal power calculation. If recruitment, labels, or eligible intervention sessions fall substantially below target, the study will be framed as feasibility work and intervention estimates will be treated as exploratory.

5.2 Participants and Recruitment

Eligible participants are adults aged 18 years or older who use Chrome or a Chromium-based browser on a personal computer, regularly visit at least two domains they are willing to monitor, and can install a research extension for the study period. Recruitment may use a university convenience sample of students and knowledge workers. Participants are not recruited or screened on the basis of addiction, ADHD, mental health status, or other clinical categories. The paper will report recruitment source, demographics relevant to interpretation, attrition, usable-session counts, and missing-label rates without claiming representativeness.

5.3 Consent, Privacy, and Data Handling

Before installation, participants receive a consent form and field-level explanation of the extension. Consent covers configured domain hostnames, declared intention, intended duration, session timing, aggregate click, scroll, and keyboard-activity counts without key values, idle time, focus loss, activity-window summaries, post-session answers, and—during Phase 2—model version, risk estimate, random assignment, prompt display, and prompt response. The extension does not collect page text, passwords, screenshots, private messages, full browsing history, source code, keystroke values, webcam data, face identity, or emotion data.

Participants receive anonymous codes such as P01. They can pause monitoring, change monitored domains, inspect records, export their data, delete individual sessions, or delete all local data. Participant exports are stored in a private, access-controlled study directory outside the source repository. Only consented research staff can access the combined dataset. Any publication artifact will contain aggregates or de-identified excerpts rather than raw participant files. Ethics approval or institutional exemption, as applicable, must be obtained before recruitment.

5.4 Phase 1: Ten-Day Collection

After onboarding, each participant chooses monitored domains from both work/learning and mixed-use contexts. On every monitored-domain session, DriftSense asks for one of the seven neutral intention categories and optional intended duration. The participant then browses normally while aggregate session and 10-second activity-window signals are stored locally. At the session boundary, the extension asks whether the visit matched the declared intention. No mid-session risk estimate or reflective intervention is shown during Phase 1.

Participants use the collector for 10 consecutive study days. The research team checks only adherence and technical failures during collection; it does not inspect page content because none is collected. At the end of Phase 1, each participant exports the primary CSV and activity-window CSV. The exports are transferred using the consented study procedure, validated, and combined outside the repository.

5.5 Model Development and Freeze

Phase 1 days 1–7 form the development period and days 8–10 form a chronological holdout for known-user future-session performance. Model preprocessing, feature definitions, hyperparameters, and threshold rules are selected using days 1–7 only, with participant-grouped folds inside that development set. The untouched days 8–10 are evaluated once. A participant-grouped or leave-one-participant-out analysis over Phase 1 is reported separately as an estimate of performance for unseen users; it is not conflated with the same-participant chronological result.

The primary deployed candidate is intention-plus-activity logistic regression. It is compared with a time threshold, domain rule, intention-only logistic regression, and activity-only logistic regression; Random Forest is an optional offline robustness model. Prediction metrics are accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Performance is computed at one, three, and five minutes and for the full session, but the intervention uses only the three-minute specification. Any session feature used at a cutoff is derived only from data observed by that time.

After holdout evaluation, one three-minute model, its preprocessing parameters, probability threshold, and maximum prompt rate are frozen. The threshold is chosen from development data to balance sensitivity against false prompts and expected burden; it is not tuned using Phase 2 outcomes. If the model performs near chance or produces an impractical eligible-session rate, the intervention may still be run as a technical feasibility pilot, but the paper will not claim successful prediction.

5.6 Phase 2: Seven-Day Micro-Randomized Intervention

The same participants use the intervention build for seven study days. Intention collection, passive signals, session boundaries, and post-session reflection remain identical to Phase 1. A session becomes randomization-eligible only when it remains active for three minutes, has the features required by the frozen model, exceeds the frozen threshold, and has not been suppressed by the prespecified prompt cap. The extension uses local random assignment with equal probability:

- (1) **Reflective-prompt assignment.** Show *Still here for your original reason?* with *Continue intentionally* and *Finish now* actions.
- (2) **Silent-control assignment.** Show no mid-session interface.

Assignment is logged before the prompt is rendered. Analysis follows assignment regardless of whether the participant interacts with the prompt. A maximum of one reflective prompt per session and a prespecified daily cap of three displayed prompts limit burden. Sessions below threshold, ending before three minutes, missing required features, or suppressed after the cap are retained for descriptive reporting but excluded from the randomized prompt-effect comparison. Both randomized assignments receive the same post-session reflection, preventing differences in label wording from defining the outcome.

5.7 Measures

The primary prediction outcomes are F1-score and ROC-AUC for intention-plus-activity prediction relative to time and domain baselines. Accuracy, precision, recall, confusion matrices, class balance, calibration where feasible, and the expected intervention rate at the frozen threshold are also reported. Results are shown separately for chronological same-participant prediction and participant-held-out prediction.

The primary intervention outcome is the proportion of complete binary-labeled eligible sessions reported as drift, compared by randomized assignment. Secondary behavioral outcomes are total session duration, duration after the three-minute assignment point, immediate session ending after a prompt, prompt response, prompt frequency, and post-session label completion. Phase 1 versus Phase 2 changes in drift rate and duration are descriptive secondary results only.

At the end of Phase 2, participants rate five statements on a five-point agreement scale: the prompts appeared at appropriate times; the prompts helped them reconsider their intention; the prompts were annoying; the extension was easy to use; and they would continue using it. One open-ended question asks what should change about prompt wording or timing. These measures assess acceptability and burden, not clinical benefit or objective attention.

5.8 Data Processing and Quality Checks

The preprocessing pipeline validates the exact participant CSV schema, confirms that each filename matches its participant identifier, rejects duplicate session identifiers and overlapping sessions, and reports incomplete or non-binary outcomes. Original participant exports are never modified. Binary analyses include only *Yes, it matched* (0) and *No, I drifted* (1); missing, *Continue intentionally*, and *Save for later* outcomes remain separate.

Session exports are joined to activity windows and intervention logs by `session_id`. Categorical features are one-hot encoded, and numeric features are standardized within training data for logistic regression. All preprocessing parameters are fitted on training partitions and applied unchanged to validation or test partitions. Quality reporting includes recruited and completed participants, sessions per participant, class distribution, label completion, files rejected, windows missing, intervention eligibility, assignment balance, prompt suppression, and attrition.

5.9 Analysis Plan

For RQ1, model performance is summarized across the predefined splits with 95% confidence intervals obtained by resampling participants where feasible. The main comparison asks whether intention-plus-activity logistic regression improves F1-score and ROC-AUC over time and domain baselines. Intention-only and activity-only models show

the incremental contribution of each feature group. A result is not interpreted as generalization to new users unless participant-held-out performance supports that claim.

For RQ2, analysis is restricted to randomized eligible sessions with binary post-session labels and follows assigned condition. For each participant with observations in both assignments, drift proportions are calculated separately for reflective-prompt and silent-control sessions. The prespecified simple comparison is a paired Wilcoxon signed-rank test on participant-level drift proportions, accompanied by the paired difference and a 95% participant-bootstrap confidence interval. Assignment-specific missing-label rates are reported. If the number of eligible sessions supports it, a mixed-effects logistic regression with a participant random intercept and prompt assignment as the main fixed effect is added as a sensitivity analysis.

Secondary outcomes are analyzed descriptively or with paired participant-level comparisons. Open-ended responses are summarized with a transparent lightweight content analysis. No individual mechanism is declared effective from a Phase 1–Phase 2 before–after difference alone. Results will be described as short-term and preliminary; a seven-day intervention cannot establish durable behavior change. No results, significance values, or effect sizes will be inserted until the real consented dataset has been analyzed.

5.10 Reproducibility and Reporting Commitments

Before recruitment, the study materials will specify the session boundary rules, seven intention options, reflection wording, binary-label mapping, prompt cap, exclusion rules, and the distinction between confirmatory and exploratory outcomes. Before Phase 2, the extension build will record a versioned model artifact, preprocessing specification, decision threshold, and randomization procedure. The analysis will produce a participant-flow summary and an accountable session-flow summary from all recorded sessions to the final RQ1 and RQ2 analysis sets. Counts will be reported for sessions ending before three minutes, missing reflection labels, incomplete activity windows, threshold eligibility, suppression by the daily cap, randomized assignment, and prompt delivery failures.

Code, schemas, isolated non-participant test fixtures, and analysis scripts may be released after review, but raw participant exports will not be placed in the public repository. A de-identified derived dataset will be shared only if consent, ethics approval, and disclosure-risk review permit it. Otherwise, reproducibility will rely on field-level documentation, deterministic preprocessing, fixed random seeds, environment information, and aggregate or disclosure-safe outputs.

5.11 Anticipated Validity Boundaries

The primary construct is a participant’s post-session judgment of alignment, not an objective measurement of attention. Labels may vary across people and may be influenced by recall, social desirability, repeated prompting, or the intervention itself. Reporting label-completion and non-binary outcomes by assignment is therefore part of the main validity assessment rather than a housekeeping detail.

The causal estimate for RQ2 applies only to sessions that meet the frozen model’s eligibility rule and are not suppressed by the prompt cap. It does not estimate the effect of prompting every browser session or every user. Randomization protects the prompt-versus-silent comparison within this eligible set, while the fixed phase order prevents a causal interpretation of the overall Phase 1–Phase 2 change. A 15–20-person convenience sample also limits precision and generalization, particularly for participant-held-out prediction. Finally, a seven-day intervention can establish short-term feasibility and preliminary efficacy only; it cannot demonstrate persistence after the extension is removed. These boundaries will remain explicit even if the observed effects are favorable.

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